

CROSS-DATASET GENERALISATION OF DEEP LEARNING FOR MALARIA CLASSIFICATION IN MICROSCOPY IMAGES

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TECHNICAL REPORT

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ABSTRACT

Convolutional neural networks (CNNs) that classify red blood cells as malaria-infected score almost perfectly on the public NIH dataset but worse on images from other laboratories. This project measures that drop and tests two suspected causes, crop format and stain colour. VGG-16, ResNet-50 and MobileNetV2 were fine-tuned on NIH cells over three seeds and tested without fine-tuning on BBBC041 and MP-IDB. Sensitivity fell by up to 63 percentage points, and the lightest network lost the most. Removing the background or matching stain colour at test time did not restore AUC. The drop is therefore not explained by these two differences, and a new laboratory will likely need recalibration or retraining.

1 INTRODUCTION

Malaria caused an estimated 282 million cases and 610,000 deaths in 2024 (World Health Organization, 2025). Diagnosis still relies on a trained microscopist reading a stained blood smear (Rajaraman et al., 2018). Software that classifies single cells could support overloaded laboratories, but only if it keeps its accuracy on smears from another laboratory, stain protocol and microscope.

The task is binary classification of single red-blood-cell images as parasitised or uninfected. Performance is measured as sensitivity and specificity at a fixed threshold, and the drop is the NIH hold-out value minus the external value. Who is affected by these errors is analysed in the ELSA report.

MRQ. To what extent do CNNs trained to classify NIH single-cell images as parasitised or uninfected lose sensitivity on BBBC041 and MP-

IDB, and specificity on BBBC041, relative to a patient-grouped NIH hold-out set, and which dataset differences explain this drop?

RQ1. How large is the drop from the patient-grouped NIH hold-out set, in percentage points, for MobileNetV2, ResNet-50 and VGG-16 applied without fine-tuning to BBBC041 and MP-IDB, and does the lighter architecture lose less than the heavier ones?

RQ2. What share of the RQ1 drop is recovered when test cells are matched to NIH at test time, first in crop format (background removed) and then in stain colour (Reinhard or histogram matching)?

RQ3. Does sensitivity on the pooled different-species MP-IDB cells (n = 140) differ from *P. falciparum* (n = 1,297) by more than the resampling interval, and which BBBC041 life stages account for the largest share of missed parasites?

Section 2 summarises the literature, Section 3 the method, Section 4 the results and Section 5 the conclusions.

2 LITERATURE REVIEW

On NIH, Rajaraman et al. (2018) reached 98.6% accuracy with a cell-level split but 95.9% when patients were kept apart, so random splits inflate the in-domain baseline. Zhao et al. (2020) tested an NIH-trained VGG-16 on BBBC041, where its AUC fell from 0.994 to 0.945. Loddo et al. (2022) found light and heavy CNNs similar within NIH, yet NIH-trained models averaged only 60.61% when separating rings from later stages on MP-IDB. Hou et al. (2026), the state of the art and the protocol adopted here, used a patient-grouped split; their compact model kept 96.4% sensitivity on BBBC041 but reached only 18.0% specificity. Xu et al. (2025) showed that stain normalisation of malaria smears depends strongly on the method: cross-platform stage accuracy was 29.34% after Reinhard transfer and 48.66% with their learned BeerLaNet.

Three gaps follow. First, the drop has been reported at different operating points and for different models, so its size and its dependence on architecture are unclear. Second, crop format (NIH cells segmented onto black, external cells cut as rectangles with background) and stain colour are named as suspected causes (Hou et al., 2026; Loddo et al., 2022; Zhao et al., 2020), but no study changes one of them while keeping the model fixed. Third, none reports how much of the drop such a change recovers. This project addresses all three with three architectures under one protocol

and a stepwise test-time ablation. The full review is in `LiteratureReview_NoahTjonSienKie.pdf`.

3 METHODOLOGY

Figure 1 shows the pipeline. Implementation details are in the Methodology Overview (MO), cited by section below.

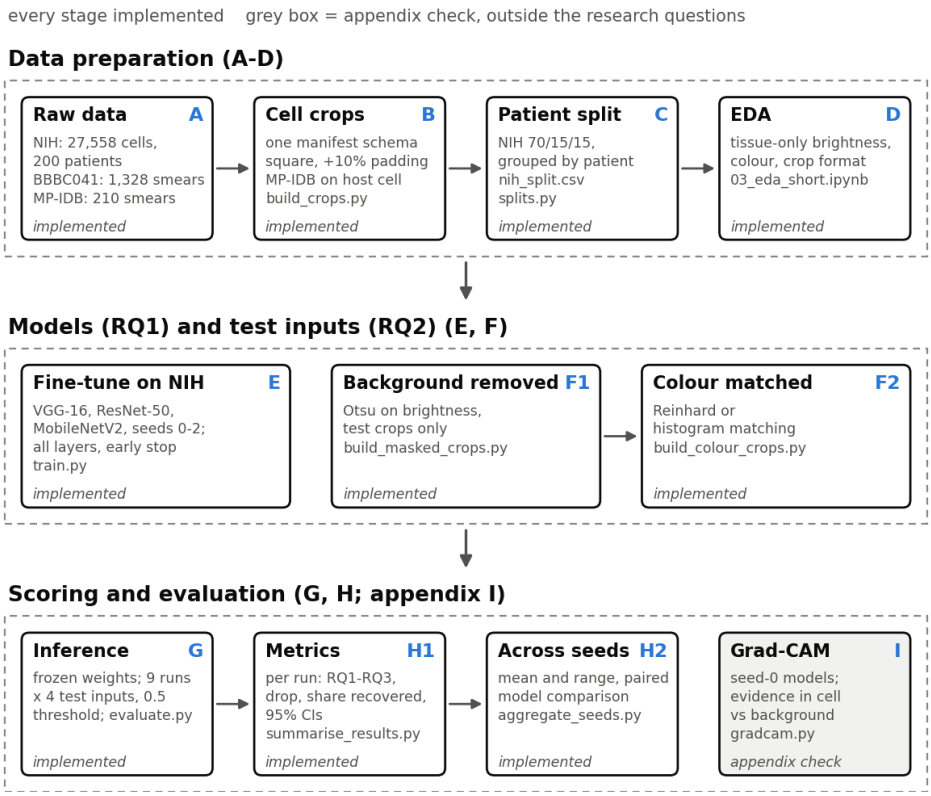


Figure 1: Pipeline. Letters match the stages in the Methodology Overview, and each box names the script that implements it. Drawn by `scripts/make_mo_figures.py`.

3.1 *Data*

NIH cell_images (Rajaraman et al., 2018) holds 27,558 crops of *P. falciparum*-infected and uninfected cells from 200 patients, segmented along the cell

outline and placed on black. It was split 70/15/15 by patient (19,203, 4,178 and 4,177 cells), so no patient's slide appears in two splits, and the hold-out set is the in-distribution reference (MO §3.2). BBBC041 (Ljosa et al., 2012) contains *P. vivax* smears from two acquisition batches, kept apart as site_a (79,569 cells, 2.7% parasitised) and site_b (5,917 cells, 5.1%); its cells were cut as squares around the annotated boxes, padded by 10%, and its 446 "difficult" cells and 103 leukocytes were excluded (MO §3.1). MP-IDB (Loddo et al., 2019) annotates only parasites, so it has no uninfected cells; each of its 1,437 parasites was recropped on its host red cell and grouped into *P. falciparum* (1,297) and the other three species (140). The exploratory analysis (MO §4) found two systematic differences: NIH crops are a median 26% black padding while no test crop has any, and BBBC041 cells absorb three to four times as much methyl-blue stain.

3.2 *Models and training*

VGG-16 (134.3 M parameters), ResNet-50 (23.5 M) and MobileNetV2 (2.2 M) were fine-tuned from ImageNet weights (He et al., 2016; Sandler et al., 2018; Simonyan & Zisserman, 2015). They are common in the malaria literature and span a sixty-fold range in size, which RQ1 needs. All layers were trained, because stained blood cells look nothing like ImageNet photographs. Each network got a single output and identical settings: Adam at a learning rate of 10^{-4}, batch size 32, flips and 90° rotations as augmentation, and early stopping on NIH validation loss (MO §5.1, Appendix Table A4). Colour augmentation was left out, so that RQ2's colour step has something to recover, and hyperparameters were fixed instead of tuned per model, so that RQ1 compares architectures and not tuning budgets (MO §5.3). Each architecture was trained with seeds 0, 1 and 2 on a Colab T4 GPU (`scripts/train.py`, `scripts/run_seeds_colab.sh`).

3.3 *Test-time ablation (RQ2)*

RQ2 edits the test crops and never the models. Step F1 removes the background: each test crop is thresholded on brightness with Otsu's method (Otsu, 1979), everything outside the cell is set to black and the crop is cut to the cell, as NIH crops were made. Otsu matched the hand-drawn NIH

outlines best of the methods tried (median Dice 0.90; MO §5.2). Step F2 then matches stain colour to NIH, either by Reinhard transfer of per-cell Lab statistics (Reinhard et al., 2001) or by histogram matching of each RGB channel, with a reference from 1,000 NIH training cells. Because the models and the cells stay fixed, any change in performance can be attributed to the step.

3.4 *Evaluation*

Sensitivity and specificity at a threshold of 0.5, fixed before testing, are the primary measures, because BBBC041 is only 2.7 to 5.1% parasitised: calling every site_a cell uninfected would score 97.3% accuracy. Specificity is measured on NIH and BBBC041 only. AUC is a threshold-free check on how well the scores rank parasitised above uninfected cells, and F1 is secondary (MO §5.5).

Share recovered is specific to this project. For a step, it is the value after the step minus the raw value, divided by the raw drop: 1 means the step closes the whole drop, 0 that it changes nothing, and a negative value that it widens the drop. It is computed only when the raw drop's 95% interval lies above zero, since dividing by a drop near zero gives meaningless values. Because sensitivity also rises when a model simply calls more cells parasitised, every share is read together with specificity and AUC for the same step.

Uncertainty has two sources. Within a run, 95% intervals come from 2,000 bootstrap resamples of whole photographs, and of patients for NIH, because cells from one photograph share its stain and focus. Across runs, results are the mean over three seeds with the lowest and highest seed in brackets. Differences in drop between models are tested with a paired bootstrap and a Bonferroni correction over 24 comparisons (MO §5.5). A finding counts as stable only if it holds in all three seeds. The NIH hold-out patients are used once, after training, and the external sets never inform training, early stopping, the colour reference or the threshold, so both are true out-of-sample tests (MO §5.5). The tables are built by `scripts/summarise_results.py`, `compare_models.py`, `aggregate_seeds.py` and `make_report_tables.py` (`make report`).

Retraining on target images and colour augmentation were rejected: both change the model and need target labels, so they would answer a different question (MO §5.7).

4 RESULTS, INTERPRETATION AND DISCUSSION

4.1 *RQ1: the drop and the architecture comparison*

Test set	Model	Sensitivity	Specificity	AUC
NIH hold-out	VGG-16	98.0 [97.6, 98.2]	97.0 [96.6, 97.5]	99.6 [99.6, 99.7]
	ResNet-50	97.3 [97.0, 97.6]	98.1 [97.7, 98.4]	99.6 [99.6, 99.7]
	MobileNetV2	97.2 [97.1, 97.5]	98.0 [97.8, 98.2]	99.6 [99.6, 99.7]
site_a	VGG-16	**2.1 [-1.1, 6.0]**	20.1 [13.2, 31.0]	**3.5 [2.5, 4.6]**
	ResNet-50	7.0 [-0.5, 17.9]	16.5 [11.0, 19.9]	5.6 [3.9, 7.6]
	MobileNetV2	21.8 [13.4, 31.9]	**5.9 [3.2, 8.1]**	5.7 [5.1, 6.2]
site_b	VGG-16	**7.1 [1.6, 11.2]**	1.3 [0.9, 1.5]	**2.2 [0.8, 3.5]**
	ResNet-50	30.4 [28.0, 33.9]	-0.6 [-1.3, 0.2]	2.9 [2.2, 3.4]
	MobileNetV2	59.4 [31.1, 80.2]	**-1.0 [-1.5, -0.7]**	8.0 [5.1, 12.3]
MP-IDB *P. falciparum*	VGG-16	**33.3 [28.6, 40.9]**	-	-
	ResNet-50	52.9 [44.6, 59.6]	-	-
	MobileNetV2	63.1 [42.4, 78.4]	-	-
MP-IDB other	VGG-16	**22.7 [18.2, 28.7]**	-	-
	ResNet-50	38.5 [35.6, 40.8]	-	-
	MobileNetV2	61.5 [54.6, 71.4]	-	-

Table 1: Model comparison: mean over seeds 0 to 2 [lowest, highest seed]. NIH hold-out rows give the value (%; AUC × 100); the other rows give the drop from that model's NIH value (pp). site_a and site_b are the two BBBC041 batches. Bold: smallest mean drop. Built by `scripts/make_report_tables.py`.

On the NIH hold-out the three models are close: sensitivity 97.2 to 98.0%, specificity 97.0 to 98.1% and AUC 0.996 (Table 1). Every external set lowers performance, but by how much depends on the model and the metric (Figure 2). VGG-16 loses the least sensitivity: 7.1 pp on site_b against 30.4 for ResNet-50 and 59.4 for MobileNetV2, and 33.3 and 22.7 pp on the two MP-IDB groups against 63.1 and 61.5 for MobileNetV2. MobileNetV2 never had the smallest sensitivity drop on any set in any seed, but it kept the most specificity on site_a in every seed (drop 5.9 pp against 20.1 for VGG-16). The answer to the second half of RQ1 is therefore no: the lighter architecture does not lose less. It calls fewer external cells parasitised, which trades missed parasites for fewer false alarms. This was not the expected outcome: a small network might have fitted less NIH-specific detail and transferred better, but the larger networks transferred their sensitivity better. The AUC drops (2.2 to 8.0 pp) are much smaller than the sensitivity drops, so much of the loss is a shift of the scores relative to the fixed threshold

rather than a loss of ranking; MobileNetV2, with the largest AUC drop on site_b (8.0 pp), also ranks cells worse.

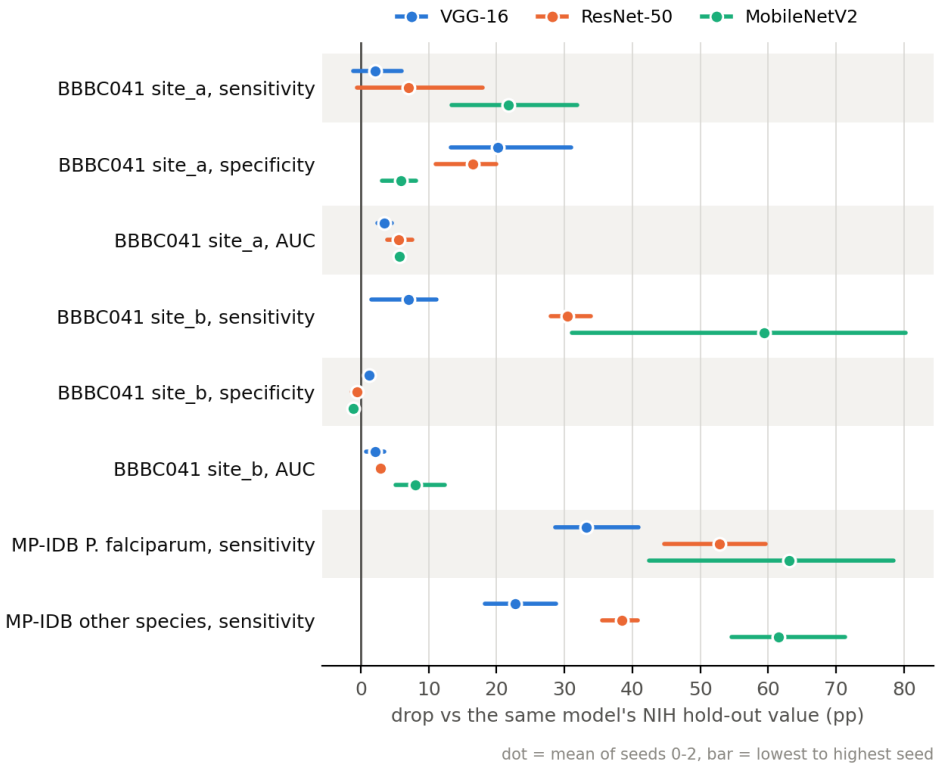


Figure 2: Drop against each model's own NIH hold-out value, per external set and metric (dot: mean of seeds 0 to 2; bar: lowest to highest seed). Built by `scripts/make_report_tables.py`.

4.2 *In-sample against out-of-sample performance*

Data	VGG-16	ResNet-50	MobileNetV2
Training (best epoch, augmented)	96.8 / 97.7	96.4 / 97.7	96.6 / 97.9
NIH validation (best epoch)	97.9 / 96.7	97.1 / 97.4	96.7 / 97.8
NIH hold-out	98.0 / 97.0	97.3 / 98.1	97.2 / 98.0
BBBC041 site_a	95.8 / 76.9	90.2 / 81.5	75.4 / 92.1
BBBC041 site_b	90.9 / 95.7	66.9 / 98.7	37.8 / 99.0
MP-IDB *P. falciparum*	64.7 / -	44.4 / -	34.1 / -
MP-IDB other species	75.2 / -	58.8 / -	35.7 / -

Table 2: Sensitivity / specificity (%) from the training data to the external sets, mean of seeds 0 to 2. Training values are measured on augmented batches at the epoch whose weights were kept. MP-IDB has no uninfected cells.

Training, validation and hold-out values differ by at most 1.2 pp for every architecture (Table 2), and early stopping kept epochs 5 to 12 (Appendix Table A3). The models did not overfit the NIH training patients; the failure appears only out of distribution, where the same models fall to 34 to 75% sensitivity on MP-IDB. A patient-grouped hold-out set therefore says little about another laboratory.

4.3 *RQ2: matching crop format and stain colour*

Removing the background recovered part of the sensitivity drop for VGG-16 and ResNet-50, for example a share of 0.31 for ResNet-50 on MP-IDB *P. falciparum* and 0.43 on site_b (Appendix Table A1). It also lowered site_a specificity for all three models, and AUC improved for no model on either BBBC041 site (Figure 3). A likely reason is two known artefacts of the masks: Otsu sometimes keeps a touching neighbour with the cell, and very dark site_a pixels fall below the black level and become padding (MO §5.2), so some uninfected cells end up with dark, parasite-like fragments on black. Reinhard transfer then lowered site_a sensitivity by a further 34 to 43 pp, while histogram matching raised sensitivity to 85 to 100% by calling almost every cell parasitised: site_a specificity fell to 4.5 to 35.3%. Its apparent recovery of 85 to 106% of the MP-IDB drops is therefore not a gain, which is why share recovered has to be read with specificity and AUC. The answer to RQ2 is that little or none of the drop is recovered when ranking is the criterion: matching crop format and colour at test time moves scores across the threshold but does not help the models separate parasitised from uninfected cells.

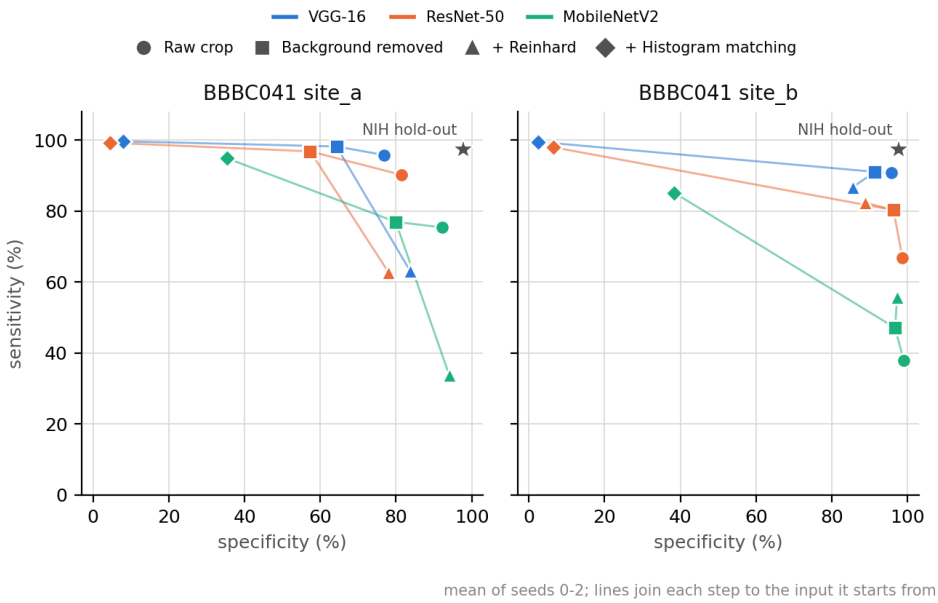


Figure 3: Sensitivity against specificity on each BBBC041 site for each test input (mean of seeds 0 to 2). The star marks the NIH hold-out. Built by `scripts/make_report_tables.py`.

4.4 *RQ3: species and life stages*

P. falciparum, the training species, was detected less often than the other species in 8 of 9 runs (Appendix Table A2). The mean gap is −10.6 pp for VGG-16 and −14.4 pp for ResNet-50, while MobileNetV2's gap changes sign between seeds. Rings account for the largest share of missed parasites on site_b for all three models (54 to 74%) and on site_a for VGG-16 and ResNet-50; for MobileNetV2 on site_a, trophozoites, the most numerous stage, account for 53%. Rings were also missed at a higher rate than trophozoites on site_a in 9 of 9 runs and on site_b in 6 of 9. Rings are small and faint, and 95% of MP-IDB *P. falciparum* parasites are rings, which plausibly explains why the training species is detected worst. A Grad-CAM check (Selvaraju et al., 2017) of the seed-0 models (Appendix Figure 4) found that the parasite received 33 to 72% of the evidence while covering 11 to 27% of the crop, for missed cells as much as for detected ones. The models look in the right place but do not recognise the parasite's appearance.

4.5 *Comparison with the literature and across seeds*

Hou et al. (2026) found 96.4% sensitivity but 18.0% specificity on BBBC041; VGG-16 shows the same pattern in a milder form, with site_a specificity down to 65.5% in its worst seed while sensitivity stayed above 91%. Zhao et al. (2020) reported an AUC of 0.945 on BBBC041 for VGG-16; here VGG-16 reached 0.961 on site_a and 0.975 on site_b. The three-architecture comparison adds how much the architecture matters under one protocol: the site_b sensitivity drop ranges from 7.1 pp for VGG-16 to 59.4 pp for MobileNetV2.

Seed variation is small on NIH and large on the external sets: the NIH values vary by at most 0.9 pp between seeds, while MobileNetV2's site_b sensitivity drop ranges from 31.1 to 80.2 pp. Of the 24 paired model comparisons, 18 point the same way in all three seeds, and 10 of those are also significant after Bonferroni correction in all three (`rq1_compare_seeds.csv`). The RQ1 ranking holds in every seed, but single-seed numbers do not: VGG-16's site_a specificity drop was 31.0 pp at seed 0 and 13.2 pp at seed 2. Reporting one training run, as most of the reviewed studies do, can misstate an external drop by a factor of two.

4.6 *Strengths and weaknesses*

The design makes the comparisons fair: one patient-grouped split, identical settings for every architecture, the same cells and threshold for every model and test input, paired intervals that resample whole photographs, and three seeds. Every table in this report is rebuilt from committed results by `make report`, so the numbers can be checked without a GPU. The weaknesses limit how far the conclusions reach. Corrections are applied only at test time, so the study cannot say whether training with colour augmentation or on masked cells would remove the drop. The fixed threshold of 0.5 mixes ranking errors with calibration errors, which the AUC only partly separates. The other-species group has only 140 cells, and the Grad-CAM check covers the seed-0 models and 100 cells per group.

5 CONCLUSION, LIMITATIONS AND FUTURE WORK

CNNs trained on NIH single cells lose much of their sensitivity on BBBC041 and MP-IDB (up to 63 pp on average) and, depending on the architecture, up to 20 pp of specificity on BBBC041, while generalising well to unseen NIH patients. The lighter MobileNetV2 loses the most sensitivity, not the least. Matching crop format and stain colour at test time moves scores across the threshold but does not improve AUC. These two differences, corrected this way, therefore do not explain the drop. Ring stages and *P. falciparum*, the training species, are missed most often. The contribution is a controlled test-time ablation that tests two commonly suspected causes, and a share-recovered measure that exposes recovery driven by the threshold alone.

The study trains on one source dataset, corrects only at test time, uses three seeds, a fixed threshold of 0.5 and only 140 non-falciparum MP-IDB cells, and Otsu masks sometimes keep a neighbouring cell. A laboratory using MobileNetV2 at site_b would miss most infected cells (sensitivity 38%), so an NIH-trained model needs local validation before any use (ELSA report). Future work should recalibrate the threshold per laboratory with a few labelled cells, train with stain augmentation or a learned normaliser such as BeerLaNet (Xu et al., 2025), and train on several source datasets with more ring-stage cells.

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A EXTENDED RESULTS

Model	Test set	Metric	Raw	Masked	+ Reinhard	+ Hist. matching	Share recovered (seeds)
VGG-16	site_a	Sensitivity	2.1	-0.3	34.7	-1.7	0.62 (2/3)
		Specificity	20.1	32.7	13.3	89.2	-0.76 (3/3)
		AUC	3.5	3.9	14.2	8.4	-0.14 (3/3)
	site_b	Sensitivity	7.1	6.9	11.2	-1.6	0.06 (2/3)
		Specificity	1.3	5.6	11.4	94.5	n/a
		AUC	2.2	3.0	5.9	24.6	-0.38 (3/3)
	MP-IDB *P. falciparum*	Sensitivity	33.3	26.3	31.4	-1.3	0.19 (3/3)
	MP-IDB other	Sensitivity	22.7	20.3	34.1	-0.4	0.11 (3/3)
ResNet-50	site_a	Sensitivity	7.0	0.4	34.5	-2.0	1.07 (2/3)
		Specificity	16.5	40.7	20.1	93.5	-1.54 (3/3)
		AUC	5.6	5.9	19.9	21.7	-0.01 (3/3)
	site_b	Sensitivity	30.4	17.0	14.9	-0.7	0.43 (3/3)
		Specificity	-0.6	1.8	9.2	91.5	n/a
		AUC	2.9	3.3	6.5	33.9	-0.11 (3/3)
	MP-IDB *P. falciparum*	Sensitivity	52.9	36.0	42.6	-1.6	0.31 (3/3)
	MP-IDB other	Sensitivity	38.5	22.0	44.7	-2.2	0.43 (3/3)
MobileNetV2	site_a	Sensitivity	21.8	20.2	63.4	2.3	0.10 (3/3)
		Specificity	5.9	18.0	3.9	62.7	-2.20 (3/3)
		AUC	5.7	11.6	14.1	16.4	-1.02 (3/3)
	site_b	Sensitivity	59.4	50.1	41.4	12.1	0.15 (3/3)
		Specificity	-1.0	1.2	0.7	59.6	n/a
		AUC	8.0	12.5	9.3	26.8	-0.50 (3/3)
	MP-IDB *P. falciparum*	Sensitivity	63.1	63.1	59.3	4.1	0.01 (3/3)
	MP-IDB other	Sensitivity	61.5	60.1	63.4	9.8	0.03 (3/3)

Table A1: RQ2: drop (pp) per test input (Masked: background removed; the colour steps start from the masked crops), mean over seeds 0 to 2, and the share of the raw drop recovered by background removal (mean over the seeds in which it is defined). Seed ranges are in `outputs/tables/report_rq2.csv`, built by `scripts/make_report_tables.py`.

Measure	n cells	VGG-16	ResNet-50	MobileNetV2
Sensitivity, *P. falciparum* (%)	1,297	64.7	44.4	34.1
Sensitivity, other species (%)	140	75.2	58.8	35.7
Gap, *P. falciparum* minus other (pp)		-10.6	-14.4	-1.6

Measure	n cells	VGG-16	ResNet-50	MobileNetV2
... seeds where *P. falciparum* detected less		3/3	3/3	2/3
site_a rings, share of all misses (%)	353	43.6	48.7	19.8
site_a trophozoites, share of all misses (%)	1,473	28.2	41.3	53.0
site_a ring miss rate (%)	353	5.2	20.7	27.1
site_a trophozoite miss rate (%)	1,473	2.4	7.0	19.5
... seeds where rings missed more than trophozoites		3/3	3/3	3/3
site_b rings, share of all misses (%)	169	73.9	61.7	53.8
site_b trophozoites, share of all misses (%)	111	17.6	32.1	39.6
site_b ring miss rate (%)	169	11.6	36.7	60.4
site_b trophozoite miss rate (%)	111	5.1	28.8	67.6
... seeds where rings missed more than trophozoites		3/3	3/3	0/3

Table A2: RQ3: sensitivity by species group, and share of misses and miss rate by BBBC041 life stage, mean over seeds 0 to 2, with the number of seeds in which each finding holds. Seed ranges are in `outputs/tables/report_rq3.csv`.

Model	Parameters (M)	Epochs run (seeds 0, 1, 2)	Kept epoch (seeds 0, 1, 2)	Min per epoch	Training time (min)
VGG-16	134.3	11, 15, 13	8, 12, 10	3.2	42 [34, 49]
ResNet-50	23.5	9, 10, 8	6, 7, 5	1.8	16 [15, 17]
MobileNetV2	2.2	9, 10, 8	6, 7, 5	1.3	12 [11, 14]

Table A3: Model size and training effort per architecture (Colab T4 GPU).

Setting	Value
Learning rate (Adam)	0.0001
Batch size	32
Maximum epochs	20
Early-stopping patience (epochs, NIH validation loss)	3
Weight decay	0.0
Input size (px)	224
Training augmentation	hflip, vflip, rot90
Decision threshold	0.5
ImageNet-pretrained, all layers trained	True

Table A4: Training settings, identical for all nine runs (read from the training logs).

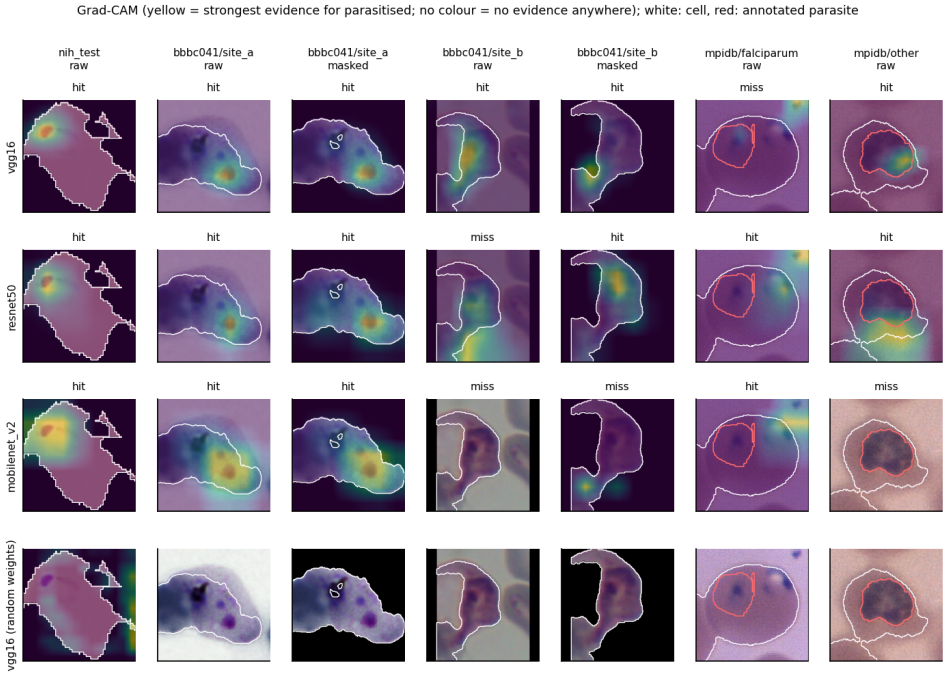


Figure 4: Grad-CAM maps of the seed-0 models on example cells (Selvaraju et al., 2017), drawn by `scripts/gradcam.py`.